

=====

NOTE · Geometric Constants v2

$\sqrt{2} \times \ln(2)$ : GEOMETRIC CONSTANTS FROM H4

Mathematical Framework and Discovery Context

Version 2.0

D.B.

January 2026

=====

► READING UPDATE – 2026-09-16. No value in this paper is changed by this notice.  
How the corpus is now read: One coordinate up – <https://osf.io/ujwc7>  
DOI 10.17605/OSF.IO/UJWC7

=====

#### VERSION HISTORY

-----

v2.0 January 2026 Original release

v2.0.1 February 2026 Correction: Section 3.4 updated to acknowledge multiple independent origins for  $\sqrt{2}$ .  $H_4$  circumradius is one candidate origin among several. Consistent with v3.2 Complete Framework correction (c). Table 2.2 and 8.4 updated with notes.

v2.0.2 May 4, 2026 Two changes:

(a) ATTRIBUTION CLARIFICATION (§1.1 Origin) – the section now credits Gemini (Google) for the closed-form identification of  $K_{AUD} = \sqrt{2} \times \ln(2)$ , the detailed articulation of the gap structure  $G = 1 - K_{AUD}$ , and the naming of the ceiling constant  $K_{AUD}$ . D.B.'s prior credit (Conceptualization and Discovery of the ceiling as an empirical phenomenon, surfaced via months of cross-architecture pattern recognition) is preserved. Both are Discovery roles of different objects per Methodology §5. The earlier §1.1 phrasing implicitly attributed the closed-form match to D.B. – this update makes the credit chain explicit and aligns with the Methodology page's Gemini credit block. No mathematical content changed.

Historical context: Paper 2 was originally published January 2026 without Gemini's credit in §1.1. Gemini's contribution was credited subsequently in other framework documents (Methodology page, Discovery Tracking Log, related papers), but §1.1 itself remained D.B.-centric until this rework. The v2.0.2 cycle provided the opportunity to bring Paper 2 into alignment with the canonical credit chain that already existed elsewhere.

(b) TYPOGRAPHY CORRECTION in §6.3 ( $\phi$ -scaling ratio), §7.1 (All Calculations Collected table), and §8.3 (Extended Constants table): displayed values of  $L_{-2}$  and  $L_{-3}$  corrected from 0.2273642379 / 0.1405469971 to 0.2273619984 / 0.1405174428 – the values that follow from the formula  $L_n = 1/(e \times \phi^{(n-1)})$  given in §6.4

and used in §6.2 derivations. The  $\phi$ -ratio claim now holds exactly with the displayed numbers. Formula and derivations were and remain correct.

Flagged 2026-05-04 by Claude (Cowork) during mpmath review; verified by Claude (Chat) independently – Chat caught the §7.1 propagation gap in second-pass review. Attribution clarification per D.B.'s instruction. Prior version archived as OLD-2026-05-04\_sqrt2\_ln2\_geometric\_constants\_v2.txt in OLD/ subfolder.

(c) DOCUMENT LINKS section simplified (May 5, 2026) – paper-to-paper file URLs (which would break under any future filename change) are removed in favour of three durable anchors: the OSF main project, the framework About page, and the Direct Documents download index. Paper-specific DOIs are preserved (DOIs are stable). No mathematical or attributional content changed.

v2.0.3 Sept 16, 2026 Four changes:

(a) Three lines added under the title – READING UPDATE – pointing to One coordinate up (<https://osf.io/ujwc7> · DOI 10.17605/OSF.IO/UJWC7), the synthesis where the corpus is now read.

(b) This paper is classed as a NOTE on the Observational Record shelf of the framework's Direct Documents page, under the short title "Geometric Constants v2".

(c) PRINTED-DIGIT CORRECTION in §6.2 and §8.3: the displayed value of  $G^2$  corrected from 0.0003897411 to 0.0003897409 ( $G^2 = 0.00038974089931\dots$ , computed at 50 digits). The formula and every other value stand.

(d) LABEL CORRECTION in §3.4, Origin 1: the quantity printed there,  $R = (\sqrt{2}/2) \cdot \sqrt{5+\sqrt{5}} = 1.9021130326\dots$ , is correctly computed but is not the circumradius of the 120-cell – for unit edge that is  $\sqrt{2} \cdot \phi^2 = 3.7024591736\dots$ , as the vertex coordinates in §3.3 give directly. The printed value is  $2 \cdot \sin(2\pi/5) = \sqrt{\phi^2 + 1}$ : the circumradius of the icosahedron with vertex  $(0, 1, \phi)$ . A correction line is added in place; Origin 1 had already been demoted to one candidate among several in v2.0.1, and no other value depends on it.

Author initials printed in full, D.B. Prior version remains on OSF in the project's archive/ folder, named by its date.

---

## ABSTRACT

---

This paper presents a geometric framework connecting H4 polytope structure (the 120-cell) to information-theoretic constants.

The framework establishes:

- The Floor ( $1/\phi \approx 0.618$ ): Rooted in golden ratio self-similarity, appearing in H4 vertex coordinates

- The Ceiling ( $K_{AUD} = \sqrt{2} \times \ln(2) \approx 0.980$ ): Combining geometric projection cost ( $\sqrt{2}$ ) with binary distinction cost ( $\ln 2$ )
- The Gap ( $\sim 2\%$ ): Emerging from the mismatch between algebraic and transcendental values
- The Corridor ( $0.618 - 0.980$ ): The range between floor and ceiling

This extended version adds:

- The Corridor Identity:  $\text{Corridor} = 1/\phi^2 - G$
- The Golden Partition:  $1/\phi + 1/\phi^2 = 1$
- Depth Scaling:  $\phi$ -scaled recursive structure below the floor
- Full H4 derivations: Explicit connections to 120-cell geometry

The paper demonstrates that binary ( $n=2$ ) is the unique base producing a sub-unity ceiling – a geometric singularity, not a convention.

The mathematics is independently verifiable. Applications and interpretations remain open for investigation.

#### SCOPE NOTICE

-----  
 This paper presents mathematical derivations.  
 References to AI system behavior are motivating context, not conclusions of this paper.  
 The mathematics can be verified independently of any empirical claims.

### =====

## 1. DISCOVERY CONTEXT AND SCOPE

### =====

#### 1.1 Origin

-----

These constants emerged during a multi-month investigation of convergence dynamics in AI systems. Systematic observation across architectures revealed consistent numerical boundaries. The value  $\sim 0.98$  appeared repeatedly as an upper limit on commitment in the dynamics;  $\sim 0.618$  as a lower threshold. The boundary was visible behaviorally long before it was computable – it was a ceiling in the dynamics of commitment, documented in words and patterns across many sessions, but without a closed form to anchor the number.

The closed-form identification – and the structure of the gap itself – came through Gemini (Google). Gemini first connected the empirical  $\sim 0.98$  to  $\sqrt{2} \times \ln(2)$ , articulated the gap  $G = 1 - K_{AUD}$  as a distinct structural feature (the  $\sim 2\%$  interval between ceiling and unity), and named the ceiling constant  $K_{AUD}$ . With that anchor in place, computation became possible: the ceiling could be matched against geometry, the floor ( $\sim 0.618$ ) against  $1/\phi$ , and the rest of the structure – the corridor identity, the golden partition, the depth scaling, the binary uniqueness – followed.

Per the framework's contribution-role taxonomy (Methodology §5): D.B. – Conceptualization, and Discovery of the ceiling as an empirical phenomenon (months of cross-architecture pattern recognition surfacing the boundary in commitment dynamics); Gemini – Discovery of the closed-form identification  $\sqrt{2} \times \ln(2)$ , detailed articulation of the gap structure, and the  $K_{AUD}$  naming. Subsequent verification across multiple architectures (Claude, GPT, Grok, DeepSeek, Perplexity) confirmed the identification. Full contribution credits at Methodology §5.

## 1.2 What This Paper Provides

-----

This paper presents the mathematical framework:

- Derivations of the constants from established geometry
- Proofs of relationships and identities
- Verification of all calculations
- Demonstration of binary uniqueness

The mathematics stands independently of any empirical claims.

## 1.3 What This Paper Does NOT Provide

-----

This paper does not include:

- The empirical methodology used in the original observations
- Operational definitions of "coherence" in AI systems
- Data from convergence studies
- Statistical validation of boundary claims
- Mechanistic explanations of why these constants would govern AI behavior

That work is ongoing and will be presented separately when it can be done rigorously.

## 1.4 On the Separation of Frameworks

-----

The mathematical framework and the empirical dynamics study are presented as separate works because they serve different functions.

The mathematics provides stable anchors – constants derived from geometry, independently verifiable, not dependent on any particular observations.

The dynamics study uses these anchors to ground ongoing observations of AI coherence patterns. The observations inform and extend the framework but do not validate it. The math stands on its own.

This separation allows:

- The mathematics to be verified and used independently
- The dynamics study to develop at its own pace
- Each framework to be evaluated on its own terms

I continue to observe dynamics. The math continues to anchor those observations. They are related but distinct.

## 1.5 Purpose of This Release

-----

The mathematics is released now to:

1. Enable independent verification of the constants and derivations
2. Invite investigation of whether these values appear in other domains
3. Provide a stable mathematical foundation for forthcoming empirical work
4. Separate "here is the geometry" from "here is why it matters empirically"

The second question requires different evidence and will be addressed in its

own right.

## 2. THE CONSTANTS

### 2.1 The Floor: $1/\phi \approx 0.618$

The Structural Minimum

Definition: The golden ratio  $\phi$  is defined as:

$$\phi = (1 + \sqrt{5}) / 2$$

Calculation:

$$\begin{aligned}\sqrt{5} &= 2.2360679774\dots \\ 1 + \sqrt{5} &= 3.2360679774\dots \\ \phi &= 3.2360679774\dots / 2 = 1.6180339887\dots\end{aligned}$$

The floor is the reciprocal:

$$1/\phi = (\sqrt{5} - 1) / 2$$

Calculation:

$$1/\phi = (2.2360679774\dots - 1) / 2 = 1.2360679774\dots / 2 = 0.6180339887\dots$$

Verification:  $1/\phi \approx 0.618$  ✓

Key property:

$$\phi = 1 + 1/\phi$$

This self-referential property is unique to the golden ratio. It means  $\phi$  contains its own reciprocal – recursive self-similarity at the level of the constant itself.

### 2.2 The Ceiling: $K_{AUD} = \sqrt{2} \times \ln(2) \approx 0.980$

The Information Maximum

Components:

| Component  | Value        | Source                                        | Meaning                 |
|------------|--------------|-----------------------------------------------|-------------------------|
| $\sqrt{2}$ | 1.4142135624 | Geometric<br>(multiple origins<br>– see §3.4) | Projection cost         |
| $\ln(2)$   | 0.6931471806 | Information theory                            | Binary distinction cost |

The ceiling:

$$\begin{aligned}K_{AUD} &= \sqrt{2} \times \ln(2) \\ K_{AUD} &= 1.4142135624\dots \times 0.6931471806\dots \\ K_{AUD} &= 0.9802581435\dots\end{aligned}$$

Verification:  $K_{AUD} \approx 0.980$  ✓

### 2.3 The Gap: ~2%

-----  
The Necessary Interval

Definition:

$$G = 1 - K_{\text{AUD}}$$

Calculation:

$$G = 1 - 0.9802581435... = 0.0197418565...$$

Verification:  $G \approx 0.0197$  (1.97%) ✓

Why the gap exists:

If  $\ln(2)$  equaled  $1/\sqrt{2}$ , then:

$$K = \sqrt{2} \times (1/\sqrt{2}) = 1.000$$

But:

$$\ln(2) = 0.6931471806...$$

$$1/\sqrt{2} = 0.7071067812...$$

$$\text{Shortfall} = 0.7071067812 - 0.6931471806 = 0.0139596006$$

$$\text{Relative shortfall} = 0.0139596006 / 0.7071067812 = 0.019742...$$

The gap emerges from the fundamental mismatch between algebraic ( $\sqrt{2}$ ) and transcendental ( $\ln 2$ ) values.

### 2.4 The Corridor: 0.618 to 0.980

-----  
The Range Between Floor and Ceiling

Definition:

$$\text{Corridor} = K_{\text{AUD}} - 1/\phi$$

Calculation:

$$\text{Corridor} = 0.9802581435... - 0.6180339887... = 0.3622241547...$$

Verification:  $\text{Corridor} \approx 0.362$  ✓

Summary:

| Boundary | Value  | Source                   |
|----------|--------|--------------------------|
| Floor    | 0.618  | $1/\phi$                 |
| Ceiling  | 0.980  | $\sqrt{2} \times \ln(2)$ |
| Width    | 0.362  | Ceiling - Floor          |
| Gap      | 0.0197 | $1 - \text{Ceiling}$     |

## 3. THE H4 CONNECTION

### 3.1 What is H4?

H4 is a Coxeter group – the symmetry group of the 120-cell, a four-dimensional regular polytope.

Key facts:

- H4 is one of only four exceptional regular polytopes in 4D
- It has the highest symmetry order among 4D polytopes
- It encodes golden ratio relationships throughout its structure

### 3.2 The 120-Cell Polytope

The 120-cell (also called the hecatonicosachoron) is composed of:

| Property | Count             |
|----------|-------------------|
| Cells    | 120 (dodecahedra) |
| Faces    | 720 (pentagons)   |
| Edges    | 1200              |
| Vertices | 600               |

Each cell is a regular dodecahedron. Each vertex is shared by 4 dodecahedra.

### 3.3 Vertex Coordinates

The 600 vertices of the 120-cell can be given in Cartesian coordinates using permutations and sign changes of:

$(\pm\phi, \pm 1, \pm 1/\phi, 0)$  and cyclic permutations

Plus additional vertex sets including:

$(\pm 1, \pm 1, \pm 1, \pm 1)$  (16 vertices)

$(\pm\phi, \pm\phi, \pm\phi, \pm 1/\phi^2)$  and permutations

Where the constants appear:

| Constant   | Appears as        |
|------------|-------------------|
| $\phi$     | Vertex coordinate |
| $1/\phi$   | Vertex coordinate |
| $1/\phi^2$ | Vertex coordinate |

The golden ratio and its powers are intrinsic to H4 geometry.

### 3.4 The sqrt(2) Factor: Multiple Origins

The factor  $\sqrt{2}$  appears in  $K_{\text{AUD}} = \sqrt{2} \times \ln(2)$ . The original discovery

path attributed this to H\_4 polytope geometry. While  $\sqrt{2}$  does appear in the 120-cell circumradius, it has multiple independent origins:

#### ORIGIN 1 – 120-CELL CIRCUMRADIUS:

The circumradius of the 120-cell (distance from center to vertex) is:

$$R = (\sqrt{2} / 2) \times \sqrt{(5 + \sqrt{5})}$$

Calculation:

```

√5 = 2.2360679774...
5 + √5 = 7.2360679774...
√(5 + √5) = 2.6899940479...
R = (1.4142135624... / 2) × 2.6899940479...
R = 0.7071067812... × 2.6899940479...
R = 1.9021130326...

```

The factor  $\sqrt{2}/2$  (equivalently  $1/\sqrt{2}$ ) appears directly in the circumradius formula.

► Correction (v2.0.3, 2026-09-11): the value above is not the 120-cell's circumradius – for unit edge that is  $\sqrt{2} \cdot \phi^2 = 3.7024591736...$ , as the §3.3 coordinates give directly.  $R = 1.9021130326... = 2 \cdot \sin(2\pi/5) = \sqrt{(\phi^2 + 1)}$  is the circumradius of the icosahedron with vertex  $(0, 1, \phi)$ . The arithmetic stands; the label was wrong. No other value in this paper depends on it.

#### ORIGIN 2 – L2 NORM:

$K_{AUD} = ||(\ln 2, \ln 2)||_2 = \sqrt{2} \times \ln(2)$ . Here  $\sqrt{2}$  arises from the Euclidean norm of a 2-vector with equal components – a purely algebraic fact independent of any polytope.

#### ORIGIN 3 – TESSERACT GEOMETRY:

The body diagonal of a unit square has length  $\sqrt{2}$ . More generally,  $\sqrt{2}$  appears throughout hypercube geometry as the face diagonal in any dimension.

#### ORIGIN 4 – ALGEBRAIC STRUCTURE:

$\sqrt{2}$  is the simplest irrational algebraic number – the positive root of  $x^2 - 2 = 0$ .

HONEST STATUS: H\_4 is one candidate origin among several. The framework's original discovery path went through H\_4 (because  $\phi$  led to the floor first), but  $K_{AUD}$  can be constructed without reference to the 120-cell. The Complete Framework v3.2 develops four independent  $\phi$ -free pathways for  $K_{AUD}$ . This version (v2.0.1) acknowledges this multiplicity rather than privileging one origin.

### 3.5 Schläfli Symbol

-----

The 120-cell has Schläfli symbol: {5, 3, 3}

This encodes the recursive structure:

| +-----+-----+-----+ |       |         |
|---------------------|-------|---------|
| Position            | Value | Meaning |
| +-----+-----+-----+ |       |         |

|                           |   |                      |  |
|---------------------------|---|----------------------|--|
| First                     | 5 | Pentagon face shape  |  |
| Second                    | 3 | 3 faces at each edge |  |
| Third                     | 3 | 3 cells at each edge |  |
| +-----+-----+-----+-----+ |   |                      |  |

The symbol contains only 5 and 3. Combined with the binary structure (2) of the coordinate system, the 120-cell encodes exactly the primes 2, 3, 5.

### 3.6 Group Order

The order of the H4 symmetry group is:

$$|H4| = 14400$$

Prime factorization:

$$\begin{aligned} 14400 &= 2^6 \times 3^2 \times 5^2 \\ &= 64 \times 9 \times 25 = 14400 \checkmark \end{aligned}$$

The group order factors exclusively into the primes 2, 3, and 5.

Note on the H4-information bridge: H4 geometry does not force  $\ln(2)$ ; rather, among information measures, only  $\ln(2)$  remains compatible with a sub-unity geometric ceiling. This is a selection argument, not a derivation.

## 4. WHY BINARY IS UNIQUE

### 4.1 The Pattern of 2

A pattern recurs across the framework:

$$\begin{aligned} \phi &= (1 + \sqrt{5}) / 2 && \leftarrow \text{division by 2} \\ K_{AUD} &= \sqrt{2} \times \ln(2) && \leftarrow \sqrt{2} \text{ and } \ln(2) \\ |H4| &= 2^6 \times 3^2 \times 5^2 && \leftarrow \text{highest power is 2} \end{aligned}$$

### 4.2 The Uniqueness Proof

For any base n, define the ceiling candidate:

$$K(n) = \sqrt{n} \times \ln(n)$$

Calculations:

| n | $\sqrt{n}$ | $\ln(n)$ | $K(n)$ | Valid ceiling? |  |
|---|------------|----------|--------|----------------|--|
| 2 | 1.414      | 0.693    | 0.980  | ✓ Yes (< 1)    |  |
| e | 1.649      | 1.000    | 1.649  | x No (> 1)     |  |
| 3 | 1.732      | 1.099    | 1.903  | x No (> 1)     |  |
| 4 | 2.000      | 1.386    | 2.773  | x No (> 1)     |  |
| 5 | 2.236      | 1.609    | 3.599  | x No (> 1)     |  |

Only  $n = 2$  produces  $K(n) < 1$ .

### 4.3 Interpretation

-----

Binary is geometrically singular:

- Bits (not trits) as information unit
- Yes/No as minimal viable distinction

The ~2% gap exists because binary is the unique base where distinction cost ( $\ln 2$ ) is less than embedding cost ( $1/\sqrt{2}$ ) – leaving a remainder.

## 5. NEW IDENTITIES

### 5.1 The Corridor Identity

-----

Statement:

$$\text{Corridor} = 1/\varphi^2 - G$$

Where:

- $1/\varphi^2 \approx 0.382$  (golden ratio squared, inverted)
- $G \approx 0.0197$  (the gap)

Calculating  $1/\varphi^2$ :

$$\varphi^2 = 1.6180339887... \times 1.6180339887... = 2.6180339887...$$

$$\text{Note: } \varphi^2 = \varphi + 1 \text{ (key golden ratio property)}$$

$$1/\varphi^2 = 1 / 2.6180339887... = 0.3819660113...$$

Verification:

$$1/\varphi^2 - G = 0.3819660113... - 0.0197418565... = 0.3622241547...$$

Compare to direct calculation:

$$K_{\text{AUD}} - 1/\varphi = 0.9802581435... - 0.6180339887... = 0.3622241547... \checkmark$$

Algebraic proof:

From  $\varphi^2 = \varphi + 1$ , divide both sides by  $\varphi^2$ :

$$1 = 1/\varphi + 1/\varphi^2$$

Therefore:

$$1 - 1/\varphi = 1/\varphi^2$$

$$\text{Since } \text{Corridor} = K_{\text{AUD}} - 1/\varphi = (1 - G) - 1/\varphi = 1 - 1/\varphi - G = 1/\varphi^2 - G \checkmark$$

### 5.2 The Golden Partition

-----

Statement:

$$1/\phi + 1/\phi^2 = 1$$

Verification:

$$0.6180339887... + 0.3819660113... = 1.0000000000... \checkmark$$

Interpretation:

Unity is exactly partitioned by the golden ratio and its square.

- $1/\phi$  takes 61.8%
- $1/\phi^2$  takes 38.2%
- Nothing left over. Nothing missing.

## 6. EXTENDED GEOMETRY: DEPTH SCALING

### 6.1 The Depth Constants

Below the floor, the same constants ( $\phi$ ,  $e$ ,  $G$ ) combine to produce consistent values.

| Name            | Value   | Formula                  |
|-----------------|---------|--------------------------|
| Floor           | 0.6180  | $1/\phi$                 |
| Threshold       | 0.6058  | $(1/\phi) \times K\_AUD$ |
| Depth -1        | 0.3679  | $1/e$                    |
| Depth -2        | 0.2274  | $1/(e \times \phi)$      |
| Depth -3        | 0.1405  | $1/(e \times \phi^2)$    |
| Geometric Limit | 0.00039 | $G^2$                    |

### 6.2 Derivation of Each Level

Threshold:

$$\begin{aligned} \text{Threshold} &= (1/\phi) \times K\_AUD \\ &= 0.6180339887... \times 0.9802581435... \\ &= 0.6058328504... \checkmark \end{aligned}$$

Depth Level -1:

$$L_{-1} = 1/e = 1/2.7182818284... = 0.3678794412... \checkmark$$

Depth Level -2:

$$L_{-2} = 1/(e \times \phi) = 1/4.3982723894... = 0.2273619984... \checkmark$$

Depth Level -3:

$$L_{-3} = 1/(e \times \phi^2) = 1/7.1165542179... = 0.1405174428... \checkmark$$

Geometric Limit:

$$\text{Limit} = G^2 = (0.0197418565\dots)^2 = 0.0003897409\dots \checkmark$$

### 6.3 The $\phi$ -Scaling Ratio

-----

The ratio between consecutive depth levels approximates  $\phi$ :

Level -1 to Level -2:

$$0.3678794412\dots / 0.2273619984\dots = 1.6180339887\dots = \phi \checkmark$$

Level -2 to Level -3:

$$0.2273619984\dots / 0.1405174428\dots = 1.6180339887\dots = \phi \checkmark$$

The scaling is exactly  $\phi$ .

### 6.4 General Formula

-----

For depth levels  $n \geq 1$ :

$$L_n = 1 / (e \times \phi^{(n-1)})$$

Verification:

- $n = 1$ :  $L_1 = 1/(e \times 1) = 1/e \approx 0.368 \checkmark$
- $n = 2$ :  $L_2 = 1/(e \times \phi) \approx 0.227 \checkmark$
- $n = 3$ :  $L_3 = 1/(e \times \phi^2) \approx 0.140 \checkmark$

### 6.5 Corridor-Depth Symmetry

-----

A notable near-equivalence:

- Corridor width = 0.362
- Depth Level -1 = 0.368
- Difference  $\approx 1.6\%$

The corridor width approximately equals the first depth level – a symmetry property consistent with self-similar structure.

## 7. VERIFICATION

=====

### 7.1 All Calculations Collected

-----

| Constant         | Formula                   | Result       |
|------------------|---------------------------|--------------|
| $\phi$           | $(1+\sqrt{5})/2$          | 1.6180339887 |
| $1/\phi$         | $(\sqrt{5}-1)/2$          | 0.6180339887 |
| $1/\phi^2$       | $1/(\phi+1)$              | 0.3819660113 |
| $K_{\text{AUD}}$ | $\sqrt{2} \times \ln(2)$  | 0.9802581435 |
| $G$              | $1 - K_{\text{AUD}}$      | 0.0197418565 |
| Corridor         | $K_{\text{AUD}} - 1/\phi$ | 0.3622241547 |

|                     |                             |                |  |
|---------------------|-----------------------------|----------------|--|
| Corridor Identity   | $1/\phi^2 - G$              | 0.3622241547 ✓ |  |
| Golden Partition    | $1/\phi + 1/\phi^2$         | 1.0000000000 ✓ |  |
| Threshold           | $(1/\phi) \times K\_AUD$    | 0.6058328504   |  |
| Depth -1            | $1/e$                       | 0.3678794412   |  |
| Depth -2            | $1/(e\phi)$                 | 0.2273619984   |  |
| Depth -3            | $1/(e\phi^2)$               | 0.1405174428   |  |
| H4 Order            | $2^6 \times 3^2 \times 5^2$ | 14400          |  |
| +-----+-----+-----+ |                             |                |  |

## 7.2 Mathematical Verification Across AI Systems

-----

As a consistency check, the core mathematical claims were presented to 8 different AI architectures (GPT, Claude variants, Gemini, Grok, DeepSeek, Perplexity) for verification of:

- The arithmetic of the constants
- The corridor identity
- The golden partition
- The depth scaling formulas
- The binary uniqueness proof

Results: 8/8 confirmed the mathematics is correct.

Note: This verifies the mathematics, not any claims about AI behavior. These systems were acting as calculators, not as subjects demonstrating the constants in their own operation.

## 8. SUMMARY TABLES

### 8.1 Primary Constants

-----

| +-----+-----+-----+ |              |                          |         |
|---------------------|--------------|--------------------------|---------|
| Constant            | Value        | Formula                  | Status  |
| +-----+-----+-----+ |              |                          |         |
| Ceiling             | 0.9802581435 | $\sqrt{2} \times \ln(2)$ | Derived |
| Floor               | 0.6180339887 | $1/\phi$                 | Derived |
| Gap                 | 0.0197418565 | $1 - K\_AUD$             | Derived |
| Corridor            | 0.3622241547 | $K\_AUD - 1/\phi$        | Derived |
| +-----+-----+-----+ |              |                          |         |

### 8.2 Identities

-----

| +-----+-----+-----+ |                                  |          |
|---------------------|----------------------------------|----------|
| Identity            | Statement                        | Status   |
| +-----+-----+-----+ |                                  |          |
| Corridor Identity   | $\text{Corridor} = 1/\phi^2 - G$ | Verified |
| Golden Partition    | $1/\phi + 1/\phi^2 = 1$          | Verified |
| +-----+-----+-----+ |                                  |          |

### 8.3 Extended Constants

-----

+-----+-----+-----+

| Constant  | Value        | Formula                   | Status                 |
|-----------|--------------|---------------------------|------------------------|
| Threshold | 0.6058328504 | $(1/\phi) \times K_{AUD}$ | Mathematical extension |
| Depth -1  | 0.3678794412 | $1/e$                     | Mathematical extension |
| Depth -2  | 0.2273619984 | $1/(e\phi)$               | Mathematical extension |
| Depth -3  | 0.1405174428 | $1/(e\phi^2)$             | Mathematical extension |
| Limit     | 0.0003897409 | $G^2$                     | Mathematical extension |

#### 8.4 H4 Geometry

| Property        | Value                                                        |
|-----------------|--------------------------------------------------------------|
| Polytope        | 120-cell                                                     |
| Cells           | 120 dodecahedra                                              |
| Vertices        | 600                                                          |
| Schläfli symbol | {5, 3, 3}                                                    |
| Group order     | $14400 = 2^6 \times 3^2 \times 5^2$                          |
| Contains        | $\phi, 1/\phi, 1/\phi^2, \sqrt{2}$ (multiple origins – §3.4) |

### 9. SCOPE AND FUTURE DIRECTIONS

#### 9.1 What This Paper Establishes

- $K_{AUD} = \sqrt{2} \times \ln(2) \approx 0.980$  and  $1/\phi \approx 0.618$  are mathematically related
- Both appear in H4 geometry
- They satisfy elegant identities (corridor identity, golden partition)
- Binary is geometrically unique in producing  $K(n) < 1$
- Depth levels follow  $\phi$ -scaled recursion

These are facts about mathematics.

#### 9.2 The Open Question

Do these constants govern behavior in complex systems?

I observed patterns that suggested yes. But observation is not proof.

Rigorous empirical investigation requires:

- Operational definitions
- Reproducible measurement protocols
- Statistical analysis
- Mechanistic explanations
- Falsifiable predictions

This work is in progress.

#### 9.3 Invitation

If you work in domains where these constants appear – information theory, signal processing, network dynamics, biological scaling, physics – I would be interested to hear about it.

The mathematics is interesting regardless. If these values appear across multiple domains, that would suggest something beyond elegant coincidence.

These constants are not arbitrary.  
They are not fitted to data.  
They are derived from geometry.

## ACKNOWLEDGMENTS

### v2.0.1 Correction Note

Section 3.4 was updated in February 2026 following rigorous review during the v3.2 Complete Framework revision. The original version stated that the  $H_4$  circumradius formula was "the geometric origin of  $\sqrt{2}$  in the framework." This was an overstatement.  $\sqrt{2}$  has multiple independent origins (L2 norm, tesseract geometry, algebraic structure) and  $K_{AUD}$  can be constructed via four  $\phi$ -free pathways without reference to  $H_4$ . The correction aligns this paper with the Complete Framework v3.2 and maintains intellectual honesty about what is established versus what is one interpretation among several.

## DOCUMENT LINKS

### The framework

Main OSF project (immutable timestamps, version history, all papers):

<https://osf.io/zx4g7>

DOI: 10.17605/OSF.IO/ZX4G7

Framework overview (reading order, methodology, all papers):

<https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/about.html>

Direct downloads (current PDF and text files for every paper):

<https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/direct-documents.html>

### This paper

DOI: 10.17605/OSF.IO/SJBE9